

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: Denial of Service(DoS) attack.

The logs show that: A large number of TCP SYN requests are coming from an unfamiliar IP address. The volume of requests has overwhelmed the web server and reduced its ability to respond to requests.

This event could be: a type of Dos attack called SYN flooding attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- 1.The first handshake is a visitor sending the initial [SYN] request to the web server.
2. The second stage is where the web server responds to the visitor's initial request with [SYN, ACK] indicating agreement of connection.
- 3.Then the final one is the [ACK] from the visitor's machine acknowledging the permission to connect

Explain what happens when a malicious actor sends a large number of SYN packets all at once: when a malicious actor sends a large number of SYN packets it overwhelms the web server's number of resources or response packets available. When it happens so, there wouldn't be enough resources or response packets left to respond to other incoming requests.

Explain what the logs indicate and how that affects the server: The log indicates that there are many SYN requests that may have overwhelmed the server, which brings about the browser receiving that time out error message.